

Zhiyuan Wen

Research Assistant Professor
Department of Computing
The Hong Kong Polytechnic University (PolyU)

Email: zyuanwen@polyu.edu.hk
Homepage: preke.github.io

EDUCATIONAL BACKGROUND

- | | |
|---------------|---|
| 2018.9-2024.1 | Ph.D. in Computer Science, The Hong Kong Polytechnic University, Hong Kong, China
Thesis: Comprehending and Reflecting Personality in Dialog Systems
Supervisor: Prof. Jiannong Cao , IEEE Fellow |
| 2021.9-2022.3 | Visiting Student at THUNLP Lab, Tsinghua University, Beijing, China
Supervisor: Prof. Maosong Sun , ACL Fellow |
| 2013.9-2017.6 | B.E. on Information Security, Sun Yat-sen University (SYSU), Guangzhou, China |

RESEARCH INTERESTS

Educational AI, Multi-agent Systems, Conversational AI, Personality and Emotion in Dialog Systems, Large Language Models

ACADEMIC SERVICE

Journal Editorship

Associate Editor, IEEE Transactions on Computational Social Systems (IEEE TCSS)

Conference Organization

Local Chair, The 12th International Conference on Behavioural and Social Computing (BESC'25)

Conference and Journal Review

Program Committee Member, The Web Conference 2025 (WWW'25)

Reviewer, ACL Rolling Review (ARR)

Reviewer, AP-Web'25, ISCC'20, IJCAI'19, AAAI'19, and IEEE BigData'19

Reviewer, ACM Transactions on Intelligent Systems and Technology

Reviewer, ACM Transactions on Information Systems

Reviewer, IEEE Transactions on Computers

Reviewer, Information Processing & Management

WORKING EXPERIENCE

2025.4-now	Research Scientist, Artificial Intelligence and Internet of Things Innovation Research Center, Nanjing Technology and Innovation Research Institute (Co-founded by Nanjing Municipal People's Government and Hong Kong Polytechnic University)
2025.4-now	Research Assistant Professor, Department of Computing, The Hong Kong Polytechnic University, Hong Kong, China
2024.2-2025.3	Postdoctoral Fellow, Department of Computing, The Hong Kong Polytechnic University, Hong Kong, China
2018.1-2018.5	Research and Development Engineer (Intern), Reliability Group, 2012 Lab, Huawei Technologies Co., Ltd., Shenzhen, China
2017.4-2017.9	Research Intern, WeiXin Group, Tencent Holdings Ltd., Guangzhou, China

TEACHING EXPERIENCE

2024 Fall	Teaching Assistant (Part-time), CCC 8015: Generative Artificial Intelligence, Lingnan University
2024 Spring	Guest Lecturer, Data Analysis based on Text Inputs 2 (Slide), COMP 5152: Advanced Data Analytics, PolyU
2021 Spring	Teaching Assistant, COMP4146: Computational Finance, PolyU
2020 Spring	Teaching Assistant, COMP1011: Programming Fundamentals, PolyU
2019 Spring	Teaching Assistant, COMP4133: Information Retrieval, PolyU
2018,19,20 Fall	Teaching Assistant, COMP3011: Design and Analysis of Algorithms, PolyU

SELECTED AWARDS AND HONORS

2023 Best Presentation Award at PolyU Research Student Conference

SELECTED PUBLICATIONS

Journal Articles

- TOIS **Zhiyuan Wen**, Jiannong Cao, Jiaxing Shen, Yang Ruosong, Shuaiqi Liu, and Sun Maosong. “Personality-affected Emotion Generation in Dialog Systems.” In: *ACM Transactions on Information Systems* (2024)
- IP&M Shuaiqi Liu, Jiannong Cao, Yicong Li, Ruosong Yang, and **Zhiyuan Wen***. “Low-resource court judgment summarization for common law systems.” In: *Information Processing & Management* 61.5 (2024). * indicates the corresponding author, p. 103796. ISSN: 0306-4573
- TKDE Shuaiqi Liu, Jiannong Cao, Zhongfen Deng, Wenting Zhao, Ruosong Yang, **Zhiyuan Wen**, and S Yu Philip. “Neural Abstractive Summarization for Long Text and Multiple Tables.” In: *IEEE Transactions on Knowledge and Data Engineering* (2023)
- IP&M **Zhiyuan Wen**, Jiannong Cao, Yu Yang, Haoli Wang, Ruosong Yang, and Shuaiqi Liu. “DesPrompt: Personality-descriptive prompt tuning for few-shot personality recognition.” In: *Information Processing & Management* 60.5 (2023), p. 103422
- IP&M Shuaiqi Liu, Jiannong Cao, Ruosong Yang, and **Zhiyuan Wen**. “Key phrase aware transformer for abstractive summarization.” In: *Information Processing & Management* 59.3 (2022), p. 102913
- WWWJ Ruosong Yang, Jiannong Cao, **Zhiyuan Wen**, and Jiaxing Shen. “Automated post scoring: evaluating posts with topics and quoted posts in online forum.” In: *World Wide Web* 25.3 (2022), pp. 1197–1221
- TSE Cuiyun Gao, Jichuan Zeng, **Zhiyuan Wen**, David Lo, Xin Xia, Irwin King, and Michael R Lyu. “Emerging app issue identification via online joint sentiment-topic tracing.” In: *IEEE Transactions on Software Engineering* 48.8 (2021), pp. 3025–3043
- TNNLS Fanghua Ye, Chuan Chen, **Zhiyuan Wen**, Zibin Zheng, Wuhui Chen, and Yuren Zhou. “Homophily preserving community detection.” In: *IEEE Transactions on Neural Networks and Learning Systems* 31.8 (2019), pp. 2903–2915

Conference Proceedings

- PerCom' 24 **Zhiyuan Wen**, Jiannong Cao, Yu Yang, Ruosong Yang, and Shuaiqi Liu. "Affective-NLI: Towards Accurate and Interpretable Personality Recognition in Conversation." In: *Proceedings of IEEE International Conference on Pervasive Computing and Communications*. IEEE, 2024
- EMNLP' 22(Findings) Shuaiqi Liu, Jiannong Cao, Ruosong Yang, and **Zhiyuan Wen**. "Long Text and Multi-Table Summarization: Dataset and Method." In: *Findings of the Association for Computational Linguistics: EMNLP 2022*. 2022, pp. 1995–2010
- IJCAI' 22 Shuaiqi Liu, Jiannong Cao, Ruosong Yang, and **Zhiyuan Wen**. "Generating a Structured Summary of Numerous Academic Papers: Dataset and Method." In: *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence*. 2023, pp. 4259–4265
- ACL' 21(Findings) **Wen, Zhiyuan**, Jiannong Cao, Ruosong Yang, Shuaiqi Liu, and Jiaxing Shen. "Automatically select emotion for response via personality-affected emotion transition." In: (2021), pp. 5010–5020
- ACL' 21(Findings) Shuaiqi Liu, Jiannong Cao, Ruosong Yang, and **Zhiyuan Wen**. "Highlight-transformer: Leveraging key phrase aware attention to improve abstractive multi-document summarization." In: *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*. 2021, pp. 5021–5027
- DASFAA' 20 Yang, Yu*, **Zhiyuan Wen***, Jiannong Cao, Jiaxing Shen, Hongzhi Yin, and Xiaofang Zhou. "EPARS: Early prediction of at-risk students with online and offline learning behaviors." In: *Database Systems for Advanced Applications: 25th International Conference, DASFAA 2020, Jeju, South Korea, September 24–27, 2020, Proceedings, Part II 25*. * indicates equal contribution. Springer. 2020, pp. 3–19
- EMNLP' 20(Findings) Ruosong Yang, Jiannong Cao, **Zhiyuan Wen**, Youzheng Wu, and Xiaodong He. "Enhancing automated essay scoring performance via cohesion measurement and combination of regression and ranking." In: *Findings of the 2020 Conference on Empirical Methods in Natural Language Processing*. 2020, pp. 1560–1569
- ICBL' 20 Yu Yang, Jiannong Cao, Jiaxing Shen, Ruosong Yang, and **Zhiyuan Wen**. "Learning analytics based on multilayer behavior fusion." In: *Blended Learning. Education in a Smart Learning Environment: 13th International Conference, ICBL 2020, Bangkok, Thailand, August 24–27, 2020, Proceedings 13*. Springer. 2020, pp. 15–24
- LREC'20 **Wen, Zhiyuan**, Jiannong Cao, Ruosong Yang, and Senzhang Wang. "Decode with

template: Content preserving sentiment transfer.” In: *Proceedings of the Twelfth Language Resources and Evaluation Conference*. 2020, pp. 4671–4679

LREC’20 Ruosong Yang, Jiannong Cao, and **Zhiyuan Wen**. “GGP: Glossary Guided Post-processing for Word Embedding Learning.” In: *Proceedings of the Twelfth Language Resources and Evaluation Conference*. 2020, pp. 4726–4730

APSEC’18 Qi Xie, **Zhiyuan Wen**, Jieming Zhu, Cuiyun Gao, and Zibin Zheng. “Detecting duplicate bug reports with convolutional neural networks.” In: *2018 25th Asia-Pacific Software Engineering Conference (APSEC)*. IEEE. 2018, pp. 416–425

Preprints

arxiv **Wen, Zhiyuan**, Jiannong Cao, Zian Wang, Beichen Guo, Ruosong Yang, and Shuaiqi Liu. “InteractiveSurvey: An LLM-based Personalized and Interactive Survey Paper Generation System.” In: *arXiv preprint arXiv:2504.08762* (2025)

arxiv **Wen, Zhiyuan**, Yu Yang, Jiannong Cao, Haoming Sun, Ruosong Yang, and Shuaiqi Liu. “Self-assessment, Exhibition, and Recognition: a Review of Personality in Large Language Models.” In: *arXiv preprint arXiv:2406.17624* (2024)

SKILLS

Programming Python, C/C++, LaTeX, Matlab

Web HTML, CSS, Javascript, Django, Flask, Bootstrap, PHP

Deep Learning PyTorch, Huggingface